

appreciation of the original shares, that appeared to generate new money. The scheme triggered an immense level of activity, with frenetic speculation in stocks and land, before it collapsed in chaos and confusion.

During the French Revolution, the history was repeated, when state paper (*assignats*) was issued against the security of confiscated land, and when overissuance produced new inflation. Drawing on the reports of French émigrés, the German poet Johann Wolfgang von Goethe added a section to his *Faust* drama in which he identified money creation with the promises of the devil. Mephistopheles persuades the emperor to issue paper money, explaining that the precise charm of the new approach to monetary security lies in the unlimited nature of note issuance, which made possible a new level of confidence in the capacity of the state: “Wise men will, when they have studied it, place infinite trust in what is infinite.” Innovation in monetary affairs thus came from the devil.

Most of the 20th century was filled with devastatingly destructive experiences with the mismanagement of currencies: inflation during war and in the aftermath of war—and in the midst of social turmoil in the 1960s and 1970s—and the deflation of the Great Depression. It took a long time for government to learn how to handle money properly.

By the late 20th century, improved monetary policymaking in most countries at last solved the problem of price stability. But this apparent monetary paradise just brought to the fore new problems. The store-of-value function looked problematic. Was it adequate to measure price stability in terms of consumer prices when there was dramatic inflation of some asset prices, in stock markets, or in real estate?

In practice, the replacement of paper currency with electronic transfers both on a wholesale level and for consumers with credit and debit cards also brought a new debate. Electronic money is convenient for making transfers, even across large distances. But it is easily trackable. Part of the demand for a new technology comes from privacy concerns: a wish to get back to the anonymity of cash transactions. In many countries, vigorous campaigns have been mounted to preserve coins and notes. Physical money represents what Fyodor

Dostoyevsky referred to in his semiautobiographical novel about convict life in Siberia, *The House of the Dead*, as “coined freedom.” Dostoyevsky was actually imagining the value of a coin to an imprisoned man, who could not spend the money to derive real resources but could dream of that freedom.

Bitcoin’s claim to combine anonymity and untraceability with security is what makes it attractive. Bitcoin originated around the time of the global financial crisis, in 2008–09. It is unclear whether the ostensible founder, the cryptically named Satoshi Nakamoto, really exists. In this

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sense Bitcoin fits perfectly into the historical pattern of diabolical currencies with a mysterious origin and uncertainty about whether trust is justified.

Bitcoin looks like a 21st century version of gold. It can be created or mined through effort. Its creators ingeniously established an analogy with gold. Just as the price of gold depended on the fact that it took a great deal of human exertion to extract it from large quantities of earth in remote locations, Bitcoin requires large amounts of computer power driven by cheap energy in remote areas of Asia or in Iceland. It marks a transformational shift in the perception of fundamental value. The metallic currencies of the premodern world encouraged the formulation of a labor theory of value: value was produced when humans added their labor to nature. Blockchain technology means that value reflects a combination of stored energy and intelligence, none of it human. It may point to a new age when most and eventually all value may be created by the nonhuman interaction of machines and energy. It is not surprising that the fear of instability—and the association of new money with diabolical qualities—has reappeared. 

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A Short History of Crypto Euphoria

An eminent economist's taxonomy of bubbles is applied to the latest financial fad

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Financial bubbles are easy to identify after the fact. But how do you spot one in advance? It's a question that has flummoxed generations of economists, not least the many who failed to foresee the global financial crisis. Now, with the rise of crypto assets, the question is taking on renewed relevance. Rather than engage in more or less informed intellectual speculation on the subject, we decided to consult the preeminent expert on bubbles: John Kenneth Galbraith.

Granted, the eminent Harvard professor and best-selling author of *The Great Crash, 1929* and *The Affluent Society* died in 2006, three years before Bitcoin came into existence. But Galbraith predicted what could happen in a caustic book, *A Short History of Financial Euphoria*, in which he analyzes major speculative events in history, from the tulip mania of 1630s Holland to the Wall Street crash of 1987, and identifies their common features. His taxonomy would have nailed the dot-com bubble and the 2008 crisis. So let's apply some of his criteria to crypto assets.

"The thought that there is something new in the world..."

"The world of finance hails the invention of the wheel over and over again, often in a slightly more unstable version," Galbraith wrote.

What does a leading crypto evangelist have to say? In an e-book titled *Bitcoin vs. the 2018 Recession*, Remy Hauxley, a self-described "cryptocurrency educator," says Bitcoin is "unlike anything we have ever seen. It'll change the world." Hauxley calls Bitcoin "a new form of gold, of money, of stocks. It's a trifecta." (He doesn't explain why a recession is so surely coming in 2018.)

"The extreme brevity of the financial memory..."

"Financial disaster is quickly forgotten," Galbraith observed. "When the same or closely similar circumstances occur again, sometimes in only a few

years, they are hailed by a new, often youthful, and always supremely confident generation as a brilliantly innovative discovery in the financial and larger economic world."

It has been about a decade since the 2008 crisis and almost a generation since the dot-com bubble, so the irrational exuberance of those two periods has largely faded from memory. Many Bitcoin traders are too young to remember either episode.

"The specious association of money and intelligence..."

Galbraith noted that people often think that "the more money, the greater the achievement and the intelligence that supports it." Rich people receive adulation for being rich, and those less well-off follow in their footsteps and make the same investments. This provides a stock of greater fools to keep the speculative engine running and, in the short term, reassures the rich that they are in fact smarter than the rest.

"Speculation became more and more intense..."

"The bulbs might now change hands several times at steadily increasing and wonderfully rewarding prices while still unseen in the ground," Galbraith wrote, describing the Dutch mania for tulips.

In the old days of bricks and mortar, an initial public offering, or IPO, was a kind of corporate rite of passage for a start-up. Nowadays, it's the initial coin offering, or ICO. Instead of stock, ICO investors buy tokens redeemable in the new currency once, *if* it goes into circulation. Unlike stock, tokens don't confer any ownership rights. Block.one, in the most successful ICO so far, has raised more than \$1.5 billion since July 2017, despite the clear indication that its tokens "do not have any rights, uses, purpose, attributes, functionalities and features." All told, ICOs raised \$6.5 billion in 2017, and more than \$4 billion in the first quarter of 2018, according to the *Wall Street Journal*.